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Non-local MRI upsampling

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ABSTRACT

In Magnetic Resonance Imaging, image resolution is limited by several factors such as hardware or time constraints. In many cases, the acquired images have to be upsampled to match a specific resolution. In such cases, image interpolation techniques have been traditionally applied. However, traditional interpolation techniques are not able to recover high frequency information of the underlying high resolution data. In this paper, a new upsampling method is proposed to recover some of this high frequency information by using a data-adaptive patch-based reconstruction in combination with a subsampling coherence constraint. The proposed method has been evaluated on synthetic and real clinical cases and compared with traditional interpolation methods. The proposed method is shown to outperform classical interpolation methods compared in terms of quantitative measures and visual observation.

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1. Introduction

In medical imaging, the image voxel size is limited by a number of factors such as imaging hardware, signal to noise ratio (SNR), time constraints or patient's comfort. In many cases, the acquired voxel size has to be decreased to fit with a concrete resolution requirement. In such situations, interpolation techniques can be applied (Thévenaz et al., 2000; Lehmann et al., 1999).

Super resolution (SR) techniques have emerged as an alternative to effectively increase the resolution of the reconstructed data (Carmi et al., 2006). In MRI, SR techniques have been previously applied to increase image resolution on functional MRI (fMRI) (Kornprobst et al., 2003) and diffusion tensor imaging (DTI) studies (Peled and Yeshurun, 2001). Unfortunately, most of these techniques are based on the acquisition of multiple low resolution (LR) images with small shifts which is time consuming and therefore not adequate for typical clinical settings.

This may be the reason because common interpolation techniques, such as linear interpolation or spline-based methods, have been used extensively in the past to decrease voxel size and increase apparent data resolution. These techniques assume that the existing points in the LR image may be represented using some kind of generic functions (e.g., *n*th order polynomials in case of B-splines) with some smoothness assumptions. However, such techniques estimate new points assuming that the existing ones (in the LR images) have the same value in the HR images which is only valid within homogeneous regions. These methods do not take into account the fact that the actual LR voxel intensity does not corresponds to an ideal sampling but a non-ideal sampling where the voxel intensity represents the weighted average of the underlying HR voxels within the LR voxel. As a result, interpolated images are typically, blurred versions of its corresponding HR reference images.

As an alternative to multiple image SR techniques, single image SR techniques perform resolution enhancement using the expected degradation model (Zhang and Cham, 2008) or exploiting the normal pattern redundancy available on image data (Ebrahimi and Vrscay, 2007; Xin Li, 2008; Protter et al., 2009; Elad and Datsenko, 2009). These latter techniques have shown very competitive results compared to classical interpolation methods. As was done for denoising, where the NL-means demonstrated high abilities to denoise images (Buades et al., 2005), these patch-based interpolation techniques enable significant improvements over the classical methods.

The exemplar-based or patch-based methods take advantage of the self-similarity of the image. In fact, instead of acquiring several

Abbreviations: NLM, non-local means; MNLM3D, multiresolution non-local means 3D; SNR, signal to noise ratio; PSNR, peak signal to noise ratio; LR, low resolution; HR, high resolution; SR, super resolution; DTI, diffusion tensor imaging.

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images of the same object to increase the image resolution, the patch-based approaches consider that is possible to extract similar information by using the non-local redundancy within a single image. As described in Ebrahimi and Vrscay (2007), while fractal-based approaches use self-similarity across scales, patch-based methods take benefit of self-similarity at the same scale. In both cases, the validity of self-similarity assumption has been verified for natural images in several domains such as compression (Fisher, 1995), denoising (Buades et al., 2005), or super resolution (Protter et al., 2009). Recently, the validity of the pattern-redundancy assumption has been also successfully verified for 2D and 3D MR images (Manjón et al., 2008; Coupé et al., 2008a,b).

As described by Ebrahimi and Vrscay (2007) and detailed by Elad and Datsenko (2009), the examples/patches can be used in several ways during image interpolation. The patches can be used to estimate the parameters that control the regularization procedure. For instance, the examples can be used to learn the prior parameters of energy functions involved in inverse problems such as Markov random field reconstruction (Roth and Black, 2005). The patches can also be used directly to reconstruct the image by averaging or best matching as in inpainting (Criminisi et al., 2004), texture synthesis (Efros and Leung, 1999) or denoising with the NL-means filter (Buades et al., 2005). More recently, hybrid methods using examples within an explicit regularization expression have been proposed (Elad and Datsenko, 2009; Xin Li, 2008). In the latter, the reconstruction obtained from examples is directly used in a regularization function within an optimization procedure with very good results.

The method proposed in this paper shares some characteristics with single image SR techniques based on self-similarity using a regularization expression (Xin Li, 2008; Protter et al., 2009). However, instead of achieving the optimization of an explicit expression using the noisy image, the proposed method is based on an iterative reconstruction–correction scheme involving a denoised version of the image. After a noise reduction stage, a patched-based non-local reconstruction is performed to take advantage of the self-similarity present in MR images. A coherence constraint is then applied in order to ensure that the reconstructed values fit with MR acquisition specificities (i.e. the fact that LR voxel intensity represents the information of a non-ideal sampling). These two steps are iteratively repeated within a coarse to fine scheme to achieve the image upsampling.

The structure of the paper is as follows. Section 2 describes the different steps of the proposed method. Section 3 proposes an extensive validation on synthetic and real MR datasets to demonstrate the ability of the proposed method to effectively increase voxel resolution in MRI by recovering high frequency information from the original LR data. To conclude, a discussion is presented in Section 4.

2. Materials and methods

In MR imaging, the common model assumes that LR voxels can be well modeled as the average of the corresponding HR voxel values plus some acquisition noise. This noise is Rician distributed in magnitude images but can be well-approximated within the imaged object as Gaussian distributed for typical clinical SNR values (SNR > 3) (Nowak, 1999; Gudbjartsson and Patz, 1995). Therefore, voxels in LR data y can be related to the corresponding x HR voxels using this expression for LR image formation:

$$y_p = \frac{1}{N}\sum_{i=1}^{N} x_i + n \tag{1}$$

where y_p is the observed LR voxel at location p, x_i is each one of the N HR voxels contained within this LR voxel and n is the additive noise from the measurement process.

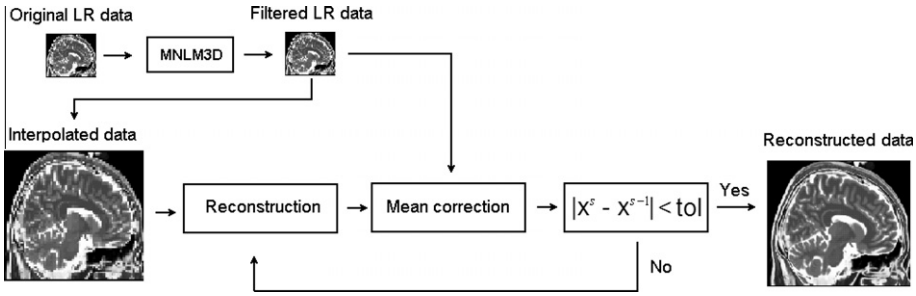


Fig. 1. Block diagram of the proposed method.

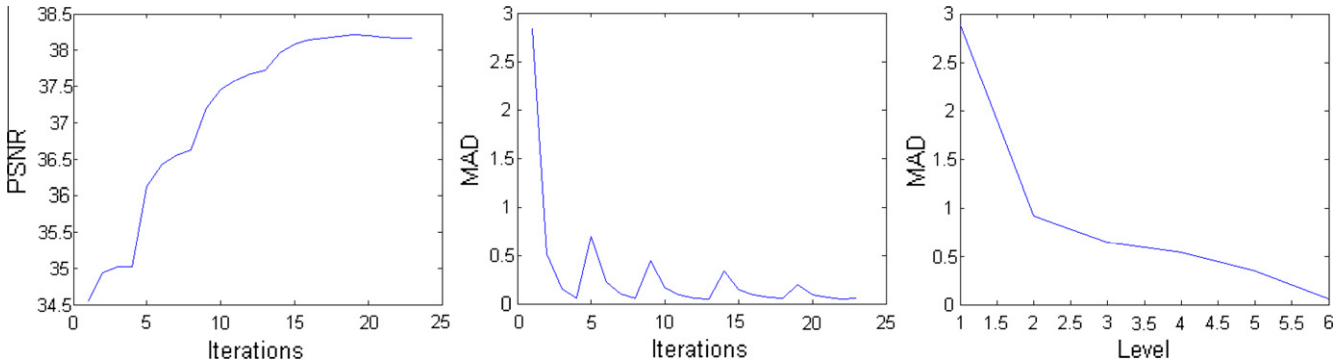


Fig. 2. On the left: evolution of the PSNR as a function of the iterations (the jumps in the PSNR values are due to the different values of h). In the center the MAD value between each two iterations is shown (for a given h value). In the right the MAD in different scales ($h = \sigma/2$ level).

In classical interpolation techniques such as linear interpolation the x_i values are calculated as a weighted average of the LR voxels y:

$$x_i = y_j \quad \text{if } (i = j) \text{ and } x_i = \sum_{j \in \Omega} w_{ij} y_j \text{ otherwise} \tag{2}$$

where the weights w_{ij} are calculated as function of the Euclidean distance between the coordinates of the new voxels and the existing LR surrounding voxels within the local area Ω. As can be noted, none of these methods takes into account the fact that existing LR voxels are the average of the underlying HR voxels and thus, these methods do not satisfy the requirements represented in the image formation model (Eq. (1)).

Therefore, the aim of any reconstruction/interpolation method should be to find the x_i values from the y_j values that accomplish the relation expressed in Eq. (1). This is a very ill-posed problem

Table 1
PSNR/SSIM values of the different methods compared for several slices thicknesses (ST).

ST (mm)	2	3	5	7	9
NN	31.33(0.9380)	27.59(0.8662)	23.93(0.7697)	22.13(0.7091)	20.85(0.6651)
Linear	34.49(0.9676)	30.17(0.9216)	25.27(0.8247)	23.07(0.7526)	21.55(0.7015)
Cubic	36.91(0.9782)	31.61(0.9366)	25.95(0.8395)	23.56(0.7658)	21.96(0.7134)
B-spline	37.80(0.9814)	32.40(0.9422)	26.06(0.8391)	23.55(0.7618)	21.94(0.7084)
Proposed	41.28(0.9900)	35.42(0.9663)	28.44(0.8708)	24.91(0.8091)	22.68(0.7384)

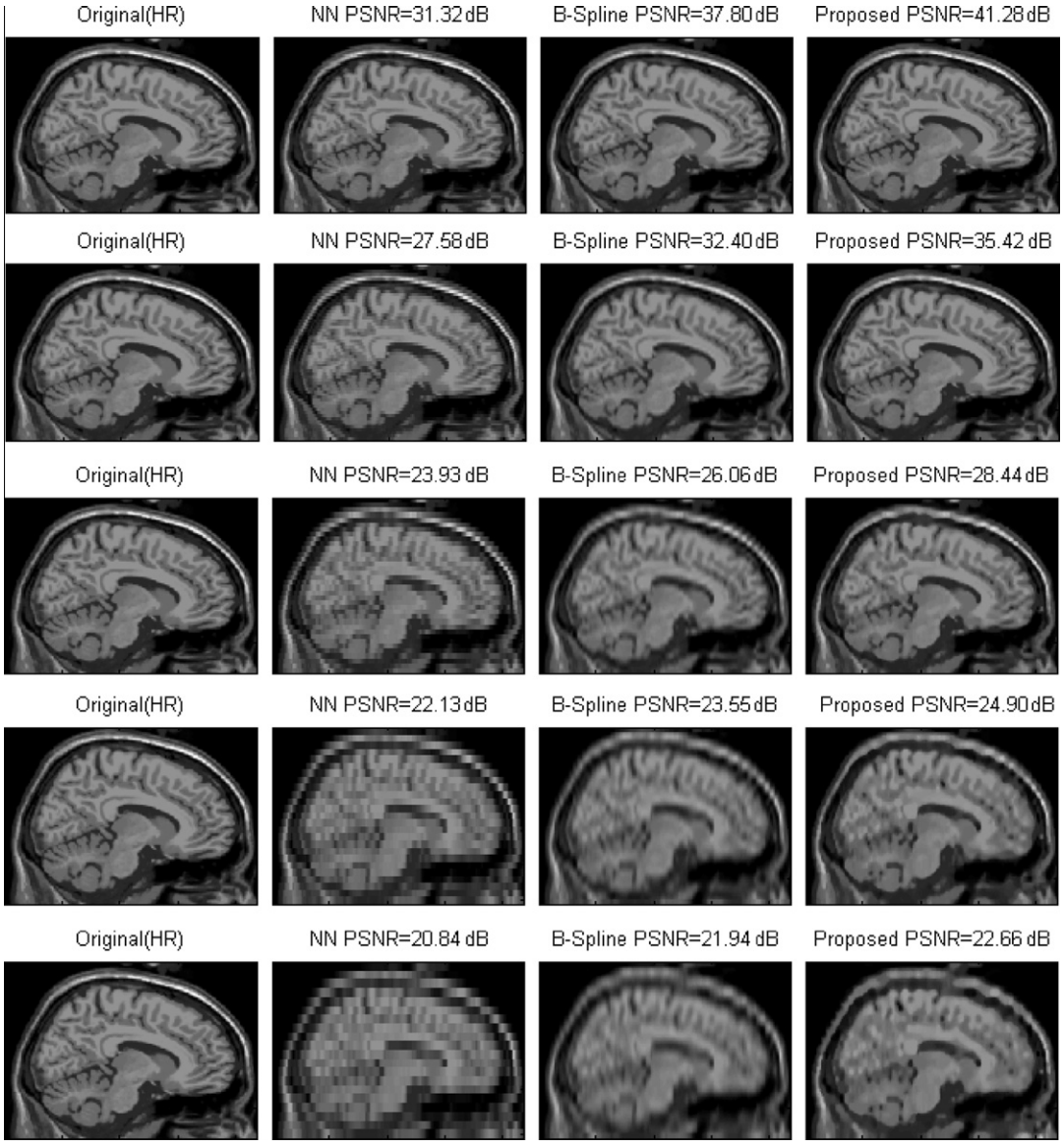


Fig. 3. Upsampling results of image datasets with different slice thicknesses: 1, 3, 5, 7 and 9 mm, from top to bottom, respectively. From left to right: Original HR data, NN interpolation, B-spline interpolation and proposed method. The proposed approach not only produces the best PSNR in all cases, but the visual appearance seems to be much better compared to other approaches.

since there are infinite x_i values that meet such condition so additional information must be included in the reconstruction process to obtain plausible results.

2.1. Proposed method

The method proposed in this paper is rooted on the application of a 3D variant of the non-local means filter. The use of this method enforces a structure-preserving constraint, rather than imposing an arbitrary smoothness constraint. In addition, as required by MR acquisition model, if the presence of noise is minimized on the LR image prior to the reconstruction process (by applying an appropriate filter), a new constraint can be defined: the downsampled version of the reconstructed image $\hat{x}$ should be exactly the same as the filtered LR image Y for all location p. Therefore, the coherence constraint can be written as:

$$Y_p - \frac{1}{N}\sum_{i=1}^{N}\widehat{x}_i = 0, \quad \forall p \in Y \tag{3}$$

This constraint has been previously applied in the SR context and referred as *subsampling consistency* (Banerjee and Jawahar, 2008). As already mentioned, there are other possibilities to solve the reconstruction process using the model described above; for example, by transforming Eq. (3) into a minimization process. However, we preferred to first denoise the image before applying an iterative reconstruction–correction scheme to avoid optimization problems such as local minima and parameter initializations.

To perform the initial denoising of the LR data, we have chosen a recently proposed robust denoising method (MNLM3D) for 3D MR images (Coupé et al., 2008b) based in the well known non-local means filter early proposed by Buades et al. (2005). It has been demonstrated that such filter is able to remove noise effectively while minimally affecting the image structure. For low SNR MR data, where the Gaussian noise approximation is no longer valid, some Rician adapted denoising methods can be used (Manjón et al., 2008; Wiest-Daesslé et al., 2008).

The proposed method uses as input the filtered LR data (as reference) and an interpolated version of the same data as an initial

Table 2
PSNR/SSIM values of the different methods compared for several slices thicknesses (ST) (noisy case, 4%).

ST (mm)	2	3	5	7	9
NN	29.81(0.8945)	26.82(0.8236)	23.57(0.7292)	21.89(0.6713)	20.68(0.6309)
Linear	31.51(0.9192)	28.64(0.8718)	24.67(0.7800)	22.71(0.7132)	21.31(0.6669)
Cubic	32.61(0.9293)	29.60(0.8857)	25.27(0.7932)	23.17(0.7246)	21.69(0.6772)
B-spline	32.93(0.9323)	30.09(0.8915)	25.38(0.7936)	23.18(0.7218)	21.69(0.6734)
Proposed	33.64(0.9376)	31.44(0.9073)	27.15(0.8223)	24.30(0.7576)	22.34(0.6944)

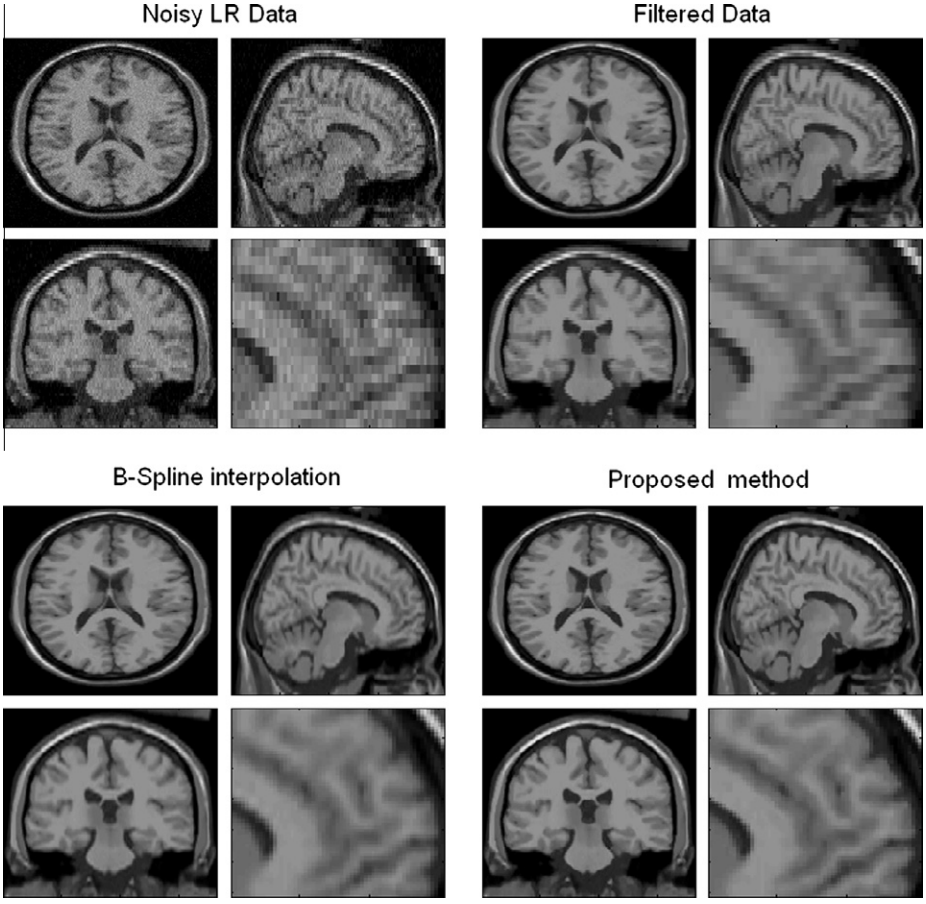


Fig. 4. Results of noisy data upsampling. The noisy (4% Gaussian noise) LR data (voxel resolution = $1 \times 1 \times 3$ mm^3) was upsampled to have a 1 mm^3 voxel resolution using the different compared methods. Both B-spline and the proposed method were applied over the filtered data.

solution for iteration 0. The proposed interpolation framework can be defined as an iterative procedure based on two steps: a patch-based non-local reconstruction, which allows very high quality reconstruction, and a mean correction step that corrects the reconstructed data in a way that the constraint expressed in Eq. (3) is satisfied.

This two step (reconstruction + correction) process is iteratively repeated within a multiscale approach (i.e. using different regularization parameters) until no significant difference is found between two consecutive estimations (defined by the mean absolute deviation (MAD) between two reconstructions at scales s and $s-1$). A block diagram of the proposed method can be observed in Fig. 1 and the details of the procedure are below.

2.1.1. Patch-based non-local reconstruction

To perform high quality image upsampling, a patch-based non-local reconstruction procedure is proposed. As shown in (Xin Li, 2008; Protter et al., 2009), the non-local means algorithm can be efficiently adapted to interpolation tasks. In fact, voxels with similar local neighborhoods can be considered as almost identical and averaged to recover local image structures. This fact has been successfully applied to perform image denoising (Buades et al., 2005) with great success. In our approach, we apply a patch-based reconstruction by using a voxel-wise 3D version of the NLM filter (Coupé et al., 2008a) with a pre-selection step adapted to the progressive reconstruction used:

$$\hat{x}_p^{t+1} = \frac{1}{C_p}\sum_{q\in\Omega} w(\hat{x}_p^t, \hat{x}_q^t)\hat{x}_q^t \tag{4}$$

$$w(\hat{x}_p^t, \hat{x}_q^t) = \begin{cases} e^{-\frac{\|N(\hat{x}_p^t)-N(\hat{x}_q^t)\|^2}{h^2}}, & \text{if } |\mu_p^t - \mu_q^t| < 3h/\sqrt{N} \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

where $\hat{x}_p^t$ is each voxel of the actual image at iteration t, w measures the similarity between $N(\hat{x}_p^t)$ and $N(\hat{x}_q^t)$ which are 3D image patches around voxels p and q, Ω is a restricted search volume surrounding the voxel being processed, C_p is a normalization constant and h is a filtering parameter related to the degree of smoothing.

As can be noted in Eq. (5), in our proposed non-local reconstruction, the pre-selection is based on the difference of the mean values μ_i^t of the $N(\hat{x}_i^t)$ patches compared at a given moment. Patches with a difference smaller than $3h/\sqrt{N}$ are used in the averaging (N is the number of voxels used in the averaging). Thus, this approach allows to perform a pre-selection based on the h parameter instead of the 3D patch local moments as suggested by Coupé et al. (2008a) which is better suited for our multiscale *coarse to fine* framework.

In this way, the voxel under study is reconstructed using all similar voxels (in the patch distance sense) in the restricted search volume Ω. Thus, the natural redundancy of information contained in the image is used to efficiently reconstruct the current voxel. Contrary to classical interpolation methods based on a straight-forward approach of modeling the data using only the local neighborhood of a voxel, the patch-based non-local approach is able to use

Table 3
PSNR/SSIM values of the different methods compared for isotropic factors.

Factor	2	3	4	5
NN	26.21(0.8100)	22.81(0.6276)	20.90(0.5222)	19.51(0.4579)
Linear	28.48(0.8817)	24.90(0.7704)	22.28(0.6708)	20.73(0.6019)
Cubic	31.33(0.9236)	26.46(0.8131)	23.55(0.7123)	21.70(0.6367)
B-spline	32.36(0.9356)	27.15(0.8293)	23.93(0.7246)	21.99(0.6452)
Proposed	35.50(0.9645)	29.65(0.8828)	26.01(0.7843)	23.39(0.6919)

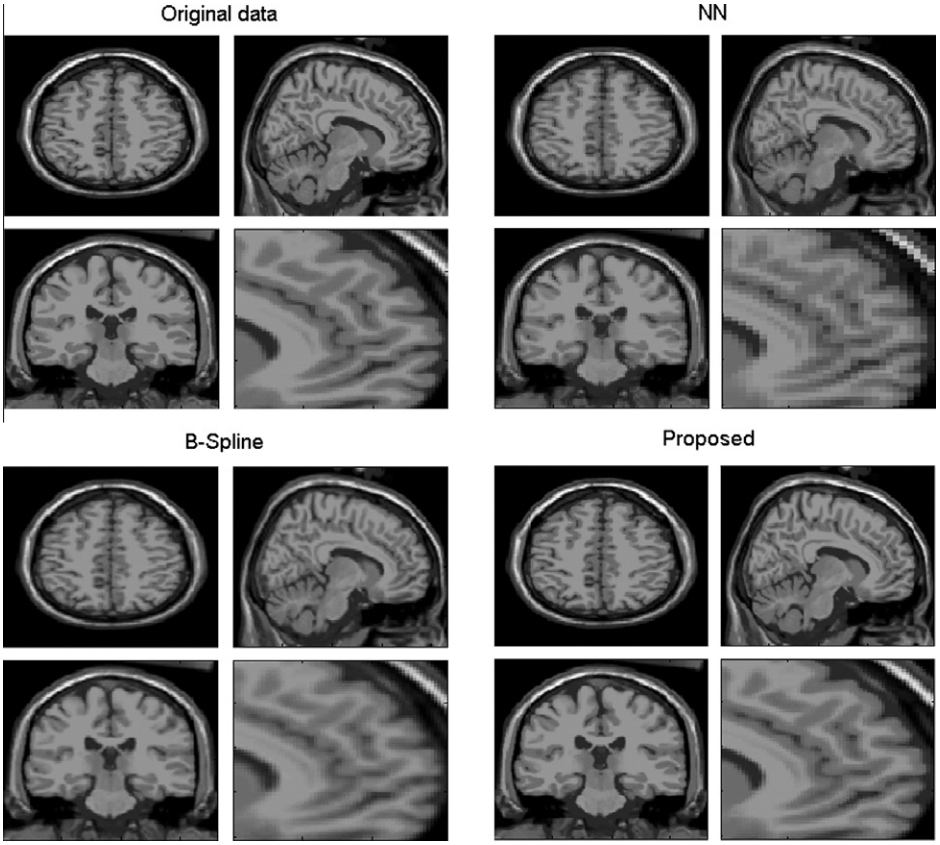


Fig. 5. Results of isotropic voxel upsampling. LR data (voxel resolution = $2\times2\times2$ mm^3) was up sampled to have a 1 mm^3 voxel resolution using the different compared methods. The proposed method produces a much more compelling result than other compared methods.

the spatial context of a voxel to use only relevant information from the search area.

2.1.2. Mean correction

As shown in by Xu et al. (2008), the non-local means filter does not ensure the mean preservation of the reconstructed value. In our case, the mean preservation of every HR reconstructed area corresponding to a given LR voxel is a critical issue since the reconstruction of $\hat{x}$ image must be accomplished under the constraint described in Eq. (3) in order to be a physically plausible solution. Thus, we proposed a mean correction step in order to ensure that the reconstructed HR image will be consistent with the original LR one.

As the downsampled version of the reconstructed HR $\hat{x}$ image must be equal to the original LR image, the local mean value of the reconstructed HR voxels is corrected to fit with the value of the original LR voxel by adding the corresponding offset.

$$\hat{x} = \hat{x} - NN(D(\hat{x}) - y) \tag{6}$$

where D is a downsampling operator that transforms actual reconstructed HR data to the original LR and NN is a nearest neighbor interpolation operation that interpolates LR data to HR.

3. Experiments and results

3.1. Implementation details

The proposed method requires an initial interpolation to start the reconstruction process. In our experiments, we used the 3rd order B-spline interpolation as implemented in MATLAB 7.4 (Mathworks Inc.).

The described patch-based non-local reconstruction scheme has three free parameters. These include the radius v of the search volume Ω, the radius f of the 3D patch used to compute the similarity among voxels and the degree of filtering h. Parameters v and f were set to 3 and 1 as suggested in Coupé et al. (2008a,b) for denoising purposes which represents $7 \times 7 \times 7$ and $3 \times 3 \times 3$ voxels 3D windows, respectively.

Regarding the h parameter, a decremental approach was used to set its value in a similar manner as previously done by Buades et al. for non-local demosaicing (Buades et al., 2009). A set of decreasing h values were used until no significant difference was found in the results between consecutive h values ($h = [\sigma/2, \sigma/4, \sigma/8, \ldots]$). Each h value is used until no significant differences are found between two consecutive iterations and then the value of h is decreased by half. This process is again repeated until no significant differences among scales (values of h) are found. In our experiments, the image difference was determined to be not significant when the MAD between iterations or scales was smaller than a given threshold $tol = 0.002\sigma$ (*being* σ the intensity standard deviation of the imaged object) which was found a good choice experimentally. In Fig. 2, an example of the evolution of peak signal to noise ratio (PSNR), and MAD of the iterative process is presented to better show the way that the proposed method works.

3.2. Simulated data evaluation

The first experiment consisted in reconstructing downsampled versions of a HR T 1-weighted (T 1w) volume of $181 \times 217 \times 180$

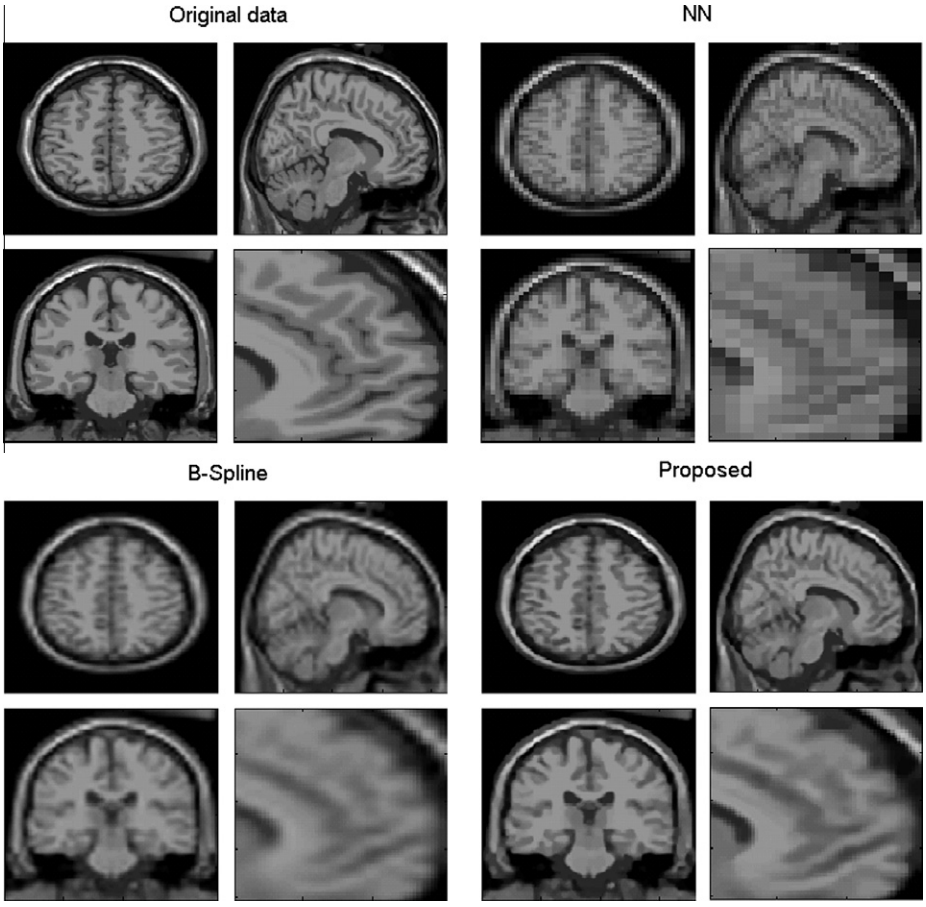


Fig. 6. Results of isotropic voxel upsampling. LR data (voxel resolution = $4 \times 4 \times 4$ mm^3) was up sampled to have a 1 mm^3 voxel resolution using the different compared methods. The proposed method produces a significantly less blurry result than other compared methods.

voxels (voxel resolution 1 mm^3) that corresponds to the HR T 1w Brainweb phantom (Kwan et al., 1999; Collins et al., 1998). This HR volume was downsampled in the z direction to simulate different slice thickness (2, 3, 5, 7 and 9 mm). These simulated LR data were then upsampled again to 1 mm^3 using different methods. The proposed method was compared to the nearest neighbor (NN), Trilinear, Cubic and B-spline (3rd order) interpolation methods as implemented in MATLAB 7.4. (Mathworks Inc). In this first experiment, no noise was added to simplify the analysis of the results (we wanted to evaluate the reconstruction process and not the denoising step). Two quality measures were used to evaluate the proposed method performance. The first was the Peak Signal to Noise Ratio (PSNR) metric which is a range invariant distance measure commonly used in image processing. The second metric was the Structural Similarity Index (SSIM) (Wang et al., 2004) which is a measure more consistent with the human visual system:

$$SSIM(x,y) = \frac{(2\mu_x\mu_y)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \quad (7)$$

where μ_x and μ_y are the mean value of images x and y, σ_x and σ_y are the standard deviation of images x and y, σ_{xy} is the covariance of x and y, $c_1 = (k_1 L)^2$ and $c_2 = (k_2 L)^2$ (L is the dynamic range, $k_1 = 0.01$ and $k_2 = 0.03$). The results can be observed in Table 1 and in Fig. 3. The proposed method obtained the best results in all the cases for both measures.

The same experiment was repeated using noisy LR data to characterize how noise and the filtering process affect the results. The LR data was corrupted by adding a Gaussian distributed noise with an amplitude similar to that obtained on real clinical datasets (4% of maximum intensity). This noisy data was then filtered using the MNLM3D method. Finally, all the upsampling methods were applied to these filtered data. Results are shown in Table 2 and Fig. 4. Again, the proposed method obtained the best results in all cases.

The next experiment consisted in reconstructing an isotropically downsampled version of the same HR T 1w volume, again to 1 mm^3 voxel resolution. This HR volume was downsampled to $2 \times 2 \times 2$, $3 \times 3 \times 3$, $4 \times 4 \times 4$ and $5 \times 5 \times 5$ mm^3 and then reconstructed again to 1 mm^3. In this experiment no noise was added. The proposed method was again compared to the NN, tri-linear, cubic and B-spline (3th order) interpolation methods. The results can be observed in Table 3 and Figs. 5 and 6. Once again, the proposed method obtained the best results in all cases.

In order to facilitate the reproducibility of our work, the source code of the described experiments and the proposed method can be found at: http://personales.upv.es/jmanjon/reconstruction/upsampling.htm.

3.3. Real clinical data evaluation

To evaluate the proposed approach on real clinical data, LR and HR T 2-w data were used. The LR dataset was obtained using a T 2-w volumetric sequence ($240 \times 240 \times 53$ voxels with a voxels resolution of $1 \times 1 \times 3$ mm^3 in a Philips Gyroscan 3 T scanner (Netherlands). The HR T 2-w data was acquired using the same sequence but using a voxel resolution of 1 mm^3 ($240 \times 240 \times 159$ voxels). Both, HR and LR data were filtered using the MNLM3D filter to remove random noise before upsampling.

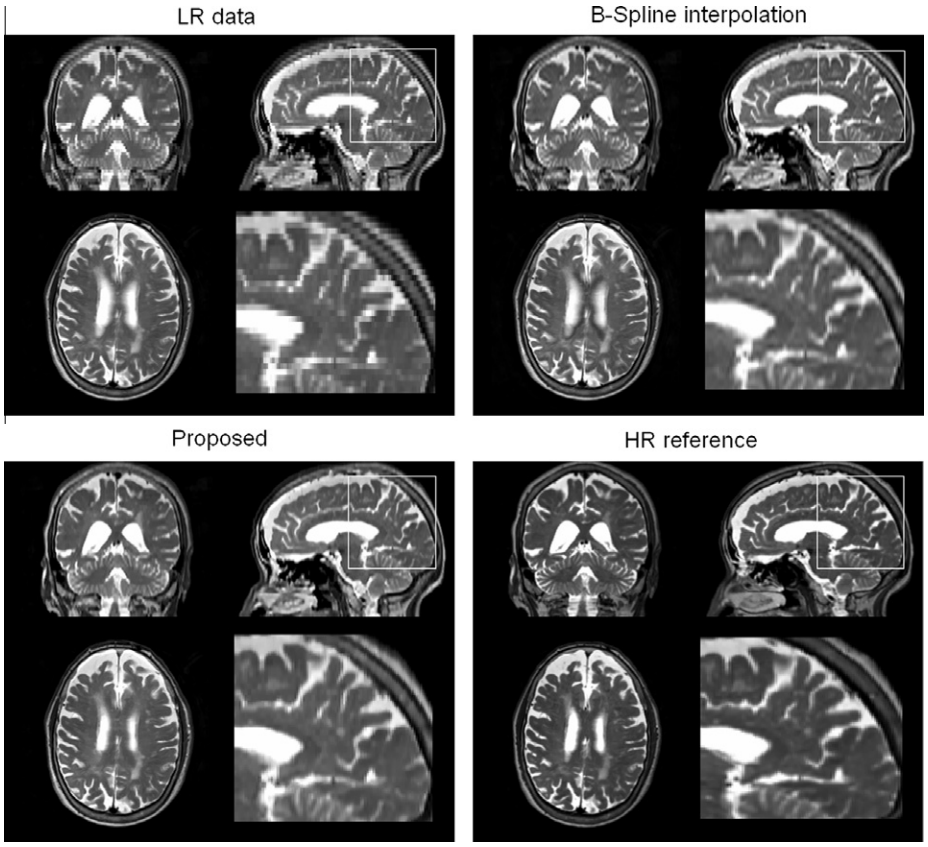


Fig. 7. Results of real clinical data upsampling. LR data (voxel resolution = $1 \times 1 \times 3$ mm^3) was upsampled to have a 1 mm^3 voxel resolution using a B-spline and the proposed method. The HR reference data is also displayed for visual comparison. Note the presence of anatomically inconsistent dark areas near the ventricles in the axial orientation when using the B-spline reconstruction. No such areas are visible when using our proposed method. A close up of posterior part of the sagittal view is displayed to better show image details.

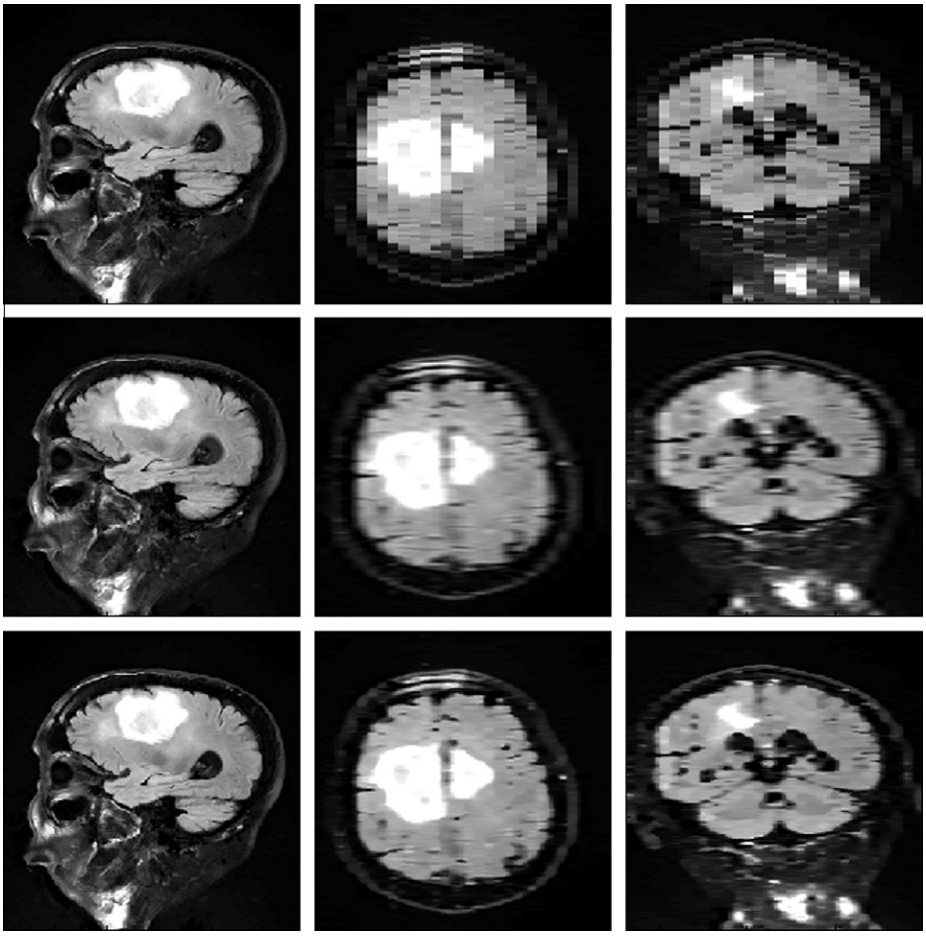


Fig. 8. Results of real clinical data upsampling. Highly anisotropic LR data (voxel resolution = $0.81 \times 0.81 \times 6\ mm^3$) was upsampled to have a $0.81 \times 0.81 \times 1\ mm^3$ voxel resolution using NN (top row), B-spline (middle row) and the proposed interpolation method (bottom row). Note that the proposed method produces less blurry images and better defined boundaries.

Two experiments were performed with this data. First a quantitative evaluation of the proposed method was done by upsampling a previously downsampled version of the HR T 2-w data ($1\ mm^3$ to $1 \times 1 \times 3\ mm^3$). In this way we were able to measure the PSNR/SSIM of the proposed method (33.53 dB/0.8263) and the reference B-spline interpolation (31.22 dB/0.7855), a commonly used technique for high quality interpolation on clinical settings. In the second experiment, the LR T 2-w data was upsampled using the B-spline interpolation and the proposed method and visually compared with the available HR T 2-w data of the same resolution. In Fig. 7, the results of this qualitative test are shown for visual comparison. As can be visually noted the proposed method was able to better reconstruct the T 2-w data in comparison with B-spline interpolation.

Additionally, the proposed methodology was tested on a pathological dataset containing a brain tumor. In this case, a LR FLAIR-w volume ($224 \times 256 \times 29$ voxels) with a voxel resolution of $0.81 \times 0.81 \times 6\ mm^3$ was used. This dataset (named CEREBRIX) was downloaded from a public MR DICOM data repository (http://pubimage.hcuge.ch:8080/). This LR data was first filtered using the MNLM3D filter and then upsampled to $0.81 \times 0.81 \times 1\ mm^3$ using the B-spline and the proposed method and the results were qualitatively evaluated. In Fig. 8, the reconstruction results using the compared methods can be observed. It can be easily seen that the upsampled volume using the proposed method is significantly less blurry and contains sharper edges, enabling a better object boundary delimitation than when using classical B-spline interpolation.

4. Discussion

A new method for high quality MR image upsampling has been presented that enables recovery of some HR data information from LR data. The proposed method has been demonstrated, using synthetic and real data, to outperform classical interpolation methods.

The improved performance of the proposed methodology can be understood considering our two main contributions. First, the proposed method uses a physically plausible model (i.e. non-ideal sampling) which has been used to constrain the iterative reconstruction process, and second, the proposed patch-based non-local reconstruction scheme prevents the occurrence of the well-known blurring effect, typically present in classical interpolation methods, by enforcing local redundant image structures. This is because, contrary to classical interpolation techniques which tend to increase the smoothness of the images in an isotropic manner, our proposed non-local reconstruction is highly anisotropic which means that each voxel is averaged only with the ones that are similar to itself (edge voxels for example). This fact, in combination with the reconstruction constraints, explains the sharpness of the results. It is clear that other reconstruction approaches could be used, as for example Total Variation (Osher et al., 2005) regularization which has less computational complexity. However, the proposed non-local reconstruction approach offers an enhanced output not only due to its edge-preserving capabilities but also due to the use of self-similar samples across the search area. Finally, it has to be noted that the initial filtering step of the proposed method may not be able to preserve small features with a contrast

similar to the noise level and therefore recovery of such features will be not possible.

As the proposed method includes the iterative application of a NLM scheme, its computational burden is high. The typical processing time of a normal MRI volume is around 1 h (it depends on the number of iterations needed to converge) using a multithreaded C implementation on Intel® Core™2Quad 2.4 GHz with 8 GB RAM. However, a number of optimizations can be performed to reduce the processing time to few minutes. For example, simply avoiding computations in the background where no useful information is obtained can reduce the computation time by approximately 50%, depending on the amount of background voxels.

The method proposed in this paper can obtain integer upsampling factors (i.e. 2×, 3×, etc.) in each dimension. We have chosen this implementation to facilitate the comprehension of the method. However, non-integer factors can be obtained upsampling the images to the closest superior integer factor and then subsampling to the desiderated resolution or by directly reconstructing the data to the final resolution by modifying Eq. (7) in a way that the constraint expressed in Eq. (3) is satisfied.

The application of the proposed methodology to upsampled quantitative MRI datasets has to be addressed clinically. Concretely, the application of this technique to upsample DTI studies could be of great interest. This and other possibilities will be addressed with further research.

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